

capabilities in three intercept scenarios, on two Banshee 80 Qinetiq target systems – a UAV purpose built to be target practice for such missiles. The Indian Air Force collaborated with Israel to develop the Medium Range Surface-to-Air Missile (MRSAM), to protect India from air attacks. Test flights to evaluate target detection and tracking have been carried out.

To engage and take down supersonic aerial targets, the DRDO is developing the Astra, a beyond visual range air-to-air missile. The missile incorporates a number of cutting edge technologies, such as electronic counter-counter measure, smokeless propulsion, and active radar guidance. The Astra, like the LSRAM, has also been properly tested against Banshee 80 targets.

BrahMos is another success story, a supersonic long range cruise missile. There are two stages to the missile, and it is one of the fastest of its kind in the world. BrahMos has a range of 290 kilometres, and operates on the fire and forget principle. BrahMos is capable of carrying a warhead of 300 kg, and has been tried out with integration on the Sukhoi Su-30MKI fighter jet.

The DRDO is also in the process of developing and inducting two anti tank missiles. Helina can be launched from a helicopter, while the Nag is designed to be launched from a special purpose infantry combat vehicle called NAMICA (Nag Missile Carrier).

VARUNA STRA

The Varunastra, named after the Hindu god of oceans, is a torpedo to take down submarines and is to be equipped on Indian warships. The torpedo can engage and neutralise submarines in shallow or deep waters. The torpedo was jointly developed by the Naval Science and Technological Laboratory (NSTL) and Bharat Dynamics. Over 95 percent of the parts used in the torpedo are indigenous. The development of the Varunastra is a major step towards making India self reliant when it comes to water defence. There are several advanced systems on board, including autonomous guiding algorithms, and a GPS recovery aid for torpedo practice. This is one of

the weapons that have recently been inducted, and the Raksha Mantri or the Defense Minister handed over the torpedo to the Indian Navy in 2016.

ARJUN MK-II

The Arjun Main Battle Tank (MBT) MK-II is an improved version of the Arjun MK-I. Due to the experience gained in developing the Arjun MK-I, the design and development of the Arjun MK-II was an accelerated process, lasting only two years. The development of the Arjun MK-I started in 1972, and the Indian Ordnance Factory only started mass producing the tank in 1996. There are a total of 84 upgrades over the Arjun MK-I. These upgrades include missile firing capabilities at longer ranges, night vision capabilities, an optional mine plough, and air defense capabilities against

known as the Kestrel, was developed in collaboration with Tata Motors. The light combat vehicle (LCV) was developed in just 18 months. The eight wheeled vehicle offers protection against small arms fire and concealed explosives. Up to 12 personnel at a time can be transported by the vehicle. The versatile vehicle can be used for a range of purposes. Depending on the requirement, it can be used to ferry personnel, a light tank, a moving command post, an ambulance, a mortar carrier, or a special purpose platform. The WhAp can also be a moving platform for chemical, biological, radiological and nuclear defense. On road, the vehicle has a maximum speed of 100 kmph, and can move at speeds of 40 kmph over terrain. There are a pair of hydrojets mounted at the back, which allows the vehicle to move



The WhAP or Kestrel designed and developed by DRDO and Tata Motors

helicopters. Improvements in the hull and turret have reduced the weight of the tank by four tons. The tank is in the advanced stage of user trials, and most of the enhancements have been already been declared successful after these tests. The systems were tested both individually, and after integration. The only major component that is yet to be tested, is an enhanced missile firing system.

WHEELED ARMOUR PLATFORM (WHAP)

Currently, the Indian Army has a fleet of old Russian light combat vehicles. There was a need to develop a new vehicle, and this was the reason that the Wheeled Armour Platform, also

through water at a speed of 10 kmph. The 25 ton vehicle is powered by a 600 horsepower diesel engine.

NAKSHATRA

The DRDO actually develops blimps, or aerostats for strategic reasons. These slow moving aircraft are ideal for surveillance purposes in sensitive areas, such as international borders. The Aerostat balloon is known as Nakshatra. A number of subsystems have been indigenously developed, apart from the balloon itself. The systems for pressure control, health monitoring, data transfer, gas management, power management and communications intelligence have been developed by the DRDO. The winch and mooring